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SEGREGATION, DISCRIMINATION AND MORTALITY IN U.S. BLACKS

ANTHONY P. POLEDNAK, PhD

Theoretically, segregation and discrimination could affect mortality among blacks (African Americans) not only through their impact on social class (including the concentration of blacks in high-poverty areas) but also by influencing health care and psychosocial factors. This paper reviews evidence that variation in mortality rates for black infants and young adults living in large metropolitan areas is associated with the level of black-white segregation. A few metropolitan areas in California with relatively low levels of segregation are shown to have low black death rates and small black-white differences in mortality rates for infants and young adult males (15–44 years old). In-depth studies are needed in these areas. Longitudinal studies of potential "risk factors" for disease and death among blacks should include: personal histories of experiences with discrimination, psychological reactions to discrimination, segregation indexes, quality of life in residential areas, and social class indicators. (*Ethnicity Dis.* 1996;6:99–108)

KEY WORDS Segregation, Discrimination, Mortality

High black (African American)-white ratios of mortality rates in infants and young adults[1,2] persist, although most rates continued to decline through 1990 (Table 1). In young adults, the black-white disparity is due not only to violent causes but to natural[3] and "preventable" causes of death.[4,5] Two studies in the early 1970's[1,2] noted the dearth of national data on black-white mortality and morbidity differentials "in the context of social and economic inequalities,"[1] and the same deficiency was noted in 1990.[6] The 1992 edition of *Health, United States*[7] documented a strong association between total U.S. death rates and educational level, consistent with the findings of epidemiologic studies,[8,9] but no data were provided on black/white ratios of death rates within educational levels. However, recently published studies involving national samples have shown a higher mortality rate among infants of college-educated black vs. white parents,[10,11] and significant black-white differences in death rates among 25 to 44 years-olds after adjustment for reported family income (in six categories).[12] In another national follow-up study, adjustment not only for reported income (in seven categories) but also for six risk factors, left a residual excess mortality rate of 31% in black versus white 35 to 54 year-olds (with only one homicide among blacks).[13] Other analyses have shown the importance of social class, marital status and family size in explaining the higher death rates for blacks for some causes (including cardiovascular diseases), but not for other causes (infectious diseases, diabetes and homicide).[14,15]

Unexplained black-white mortality differences could reflect (at least in part) several hypotheses: inadequate control for black-white differences in current social class, failure to consider the effect of social class in earlier life (including childhood), and intergenerational effects of social class.[16,17] However, still other hypotheses should be considered for the black-white mortality differences. The "American apartheid"[18] system of discrimination (in such areas as housing, bank loans, and employment)[19-23] and de facto segregation[24,25]

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TABLE 1.—Time Trends in Mortality in U.S. Blacks (B) and Whites (W), 1950–90, in Selected Age Groups

	Infant Deaths per 1,000 Live Births[a]			Death Rate, Age 15–44 Years[b] (per 100,000)					
				Males			Females		
Year	B	W	B/W	B	W	B/W	B	W	B/W
1950	43.9	26.8	1.6	517	226	2.3	418	129	3.2
1960	44.3	22.9	1.9	422	202	2.1	285	102	2.8
1970	32.6	17.8	1.8	569	220	2.6	264	105	2.5
1980	21.4	11.0	1.9	402	194	2.1	165	82	2.0
1990	17.0	7.7	2.2	424	181	2.3	157	69	2.3

[a] Males and females combined.[7]

[b] Age standardization was by the "direct" method; age-specific death rates (15–24 to 35–44 years) for each race[7] were multiplied by the corresponding proportions of the total standard population (1980 U.S.) age 15–44.

could affect mortality and morbidity among blacks not only indirectly (through its impact on social class) but also directly, by influencing access to, and quality of, health care, psychosocial factors and the physical environment (Figure 1). The second possibility is the subject of this review.

Studies of segregation and mortality

In 1950 Yankauer[26] reported that the proportion of total live births that was nonwhite was positively associated with fetal and infant death rates among regions ("health areas") of New York City. Although the analyses did not control for variation in socioeconomic status among the areas, this work should have been groundbreaking. However, in the comprehensive 1989 report on the status of blacks in American society,[20] the issue of the potential health effects of segregation was only briefly mentioned and no studies were cited. In 1990 McCord and Freeman[27] showed that the highest mortality rates in New York City, especially among young adults, were in areas with a black majority population; variation in the black-white disparity in social class was not considered, but the striking findings indicated a major challenge to public health and social policy.

Other studies, however, have examined the association between mortality rates and both socioeconomic variables and segregation, in multiple regression models. Segregation was measured by the index of dissimilarity, which measures the unevenness of the residential distribution of a specific minority group across the subunits (census tracts or block groups) within an urban area; its value theoretically ranges from .000 (no segregation) to 1.000 (complete segregation), and is independent of the sizes of the populations. The formula is

$$D = .5\,|(x_i/X) - (y_i/Y)|,$$

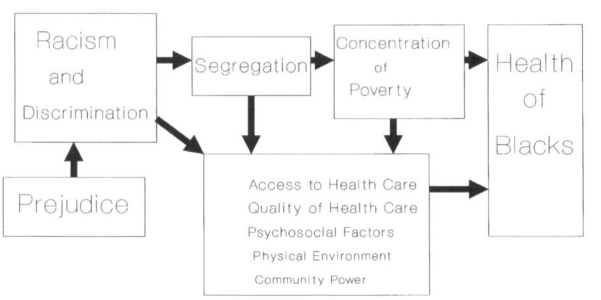

FIG 1.—Pathways whereby segregation and discrimination may influence the health status of blacks.

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TABLE 2.—Prediction of the Average Annual Infant Mortality Rate in Blacks and Whites in 38 Large U.S. Metropolitan Statistical Areas in 1987–89, by Multiple Regression

Independent variable	Dependent Variable					
	Black Rate			White Rate		
	Coeff.	t[a]	*P*	Coeffic.	t[a]	*P*
Race-specific poverty in 1989 (%)	−0.051	−0.81	.422	0.098	1.74	.091
Segregation index, 1990[b]	+0.141	3.48	.001	−0.002	−0.13	.901
Constant	9.296			7.542		
	Adjusted R^2 = .278			Adjusted R^2 = .085		

Means (and standard deviations) of infant death rates (per 1000 live births) for the 38 MSAs were 8.2 (0.9) for whites and 17.9 (2.8) for blacks.

[a] The regression coefficient divided by its standard error.

[b] The index of dissimilarity (see text for formula), multiplied by 100 (as reported by Farley and Frey).[29]

where x_i and y_i are the numbers of persons in subunit i and X and Y are the total numbers for each ethnic group in a city or metropolitan area.[25,28] Recently, indexes based on the above formula (multiplied by 100) were reported for various cities for 1980 and 1990 based on block groups (averaging 903 residents in 1980 and 564 in 1990)[24,29] rather than census tract subunits.[25]

Infant mortality

When black-white differences in poverty rate, median family income and proportion of families with a female householder were included in multiple regression models, the black-white difference in infant mortality rate (1982-86) was positively associated with segregation (i.e., index of dissimilarity in 1980)[25] among 38 U.S. metropolitan statistical areas (MSAs) with populations greater than 1 million in 1980.[30] The largest black-white differences in infant mortality were in MSAs in the Northeast and North Central/Midwest regions. The author[30] was unaware that a positive association between segregation and the infant mortality rate (1981-85) among blacks (and not whites), independent of poverty rates in multiple regression models, had also been shown among 176 U.S. cities.[31] LaVeist[31] found that this association held when the black low birthweight rate and the rate of unwed mothers were included in the model, suggesting that the association with segregation may not be due entirely to an effect on birthweight and may not be completely explained by a positive association between segregation and the rates of births to unwed black mothers.

Analysis of average annual infant mortality rates in 1987-89 in 38 large MSAs (studied previously)[30] show the persistence of the association between degree of segregation and infant mortality in blacks, independent of the black poverty rate, and the absence of an association in whites (Table 2). The black infant mortality rate was correlated (Pearson r) less strongly with the black poverty rate ($r=.167$, $p=.158$) than with the segregation index ($r=.514$, $p<.001$).

In 1911-15 in eight U.S. cities in the North and North-Central regions, within groups defined by father's income, blacks (most of whom lived in Baltimore, Maryland) had infant mortality rates that were only approximately 10% higher than native whites. Infants of low-income mothers not employed outside the home showed no black disadvantage.[32] It might be hypothesized that the rise in residential segregation in the North, after the migration of blacks to northern ghettos starting around the time of World War I,[28] led to the emergence of factors that influence black infant mortality independent of an individual's socioeco-

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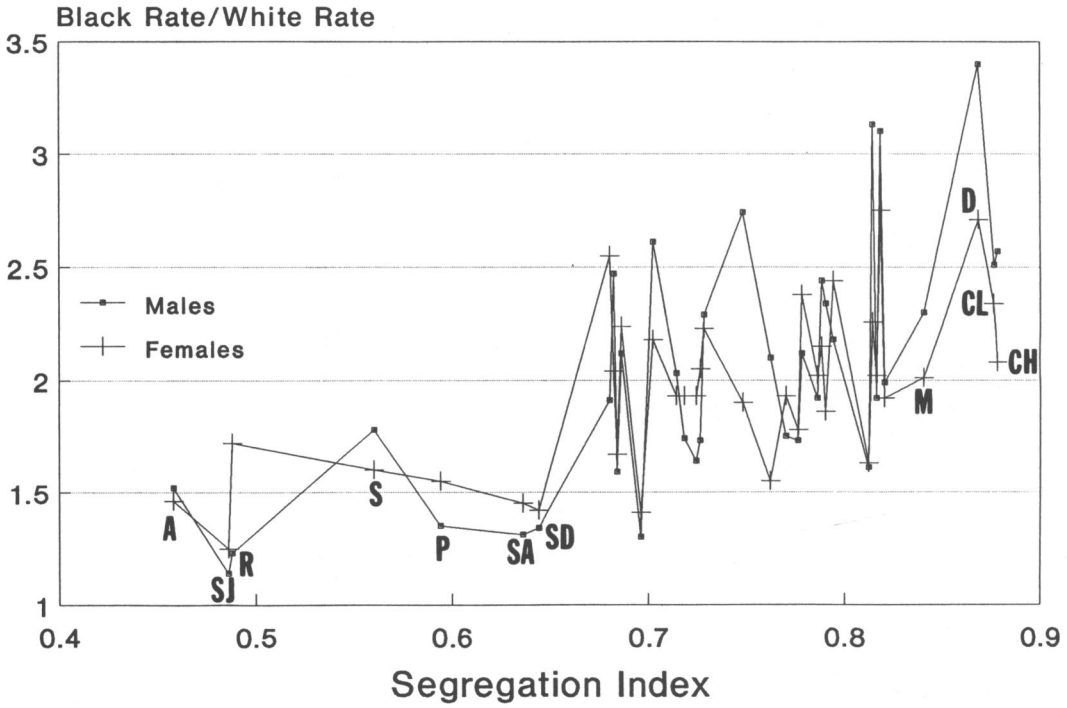

FIG 2.—Black/white ratio of age-standardized mortality rates (1982-86) in 38 large MSAs, by segregation index of MSA; drawn from data in reference 50; sgregation indexes are from reference 8. A: Anaheim CA; CH: Chicago IL; CL: Cleveland OH; D: Detroit MI; M: Minneapolis MN; P: Phoenix AR; R: Riverside CA; S: Sacramento CA; SA: San Antonio TX; SD: San Diego CA; SJ: San Jose CA.

nomic status. In 1982-86 a few California MSAs had black-white differences in infant mortality rate (unadjusted for social class) of only about 2 to 4 per 1,000 live births, despite considerable black-white differences in poverty rates.[30] In-depth studies are needed of infant mortality in these MSAs.

Mortality in young adults

Segregation (i.e., the index of dissimilarity) has been associated with the black/white ratio of age-adjusted death rates (all causes of death, 1982-86) for age 15 to 44 year-olds in 38 large MSAs;[33] in multiple regression analyses, the positive association was largely limited to blacks. The MSAs with the highest black/white mortality ratios (e.g., Detroit, Michigan, Newark, New Jersey, and St. Louis, Missouri) were in the Northeast and North-Central/Midwest regions and had some of the highest segregation ratios (Figure 2) but not the highest black/white ratios of poverty rates. While Figure 2 also represents the effect of the positive association between segregation and low social class in blacks, the segregation index[25] was a significant predictor of black/white mortality ratio when the black/white ratio of poverty rates, or when the black/white ratio of the proportion of households with low incomes (<$7500), was included in regression models.[33] Noteworthy is the finding that several MSAs in the West and Southwest had black/white mortality ratios less than 1.5, even though the lowest segregation indexes were >.45.

The largest California MSAs/PMSAs are useful in analyses of the association between segregation index and mortality rates because of the wide range of black-white

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TABLE 3.—Age-Standardized Death Rates (per 100 000) for Age 15–44 Years in Black (B) and White (W) Males in Seven Large California MSAs/PSMAs,[a] 1987–89

	Segregation Index[b]		Poverty Rate[c]			Deaths and Rates				
						Blacks		Whites		
MSA/PSMA	1980	1990	B	W	B/W	No.	Rate	No.	Rate	B/W
Anaheim	47	43	9.7	6.7	1.4	81	212	2637	177	1.20
San Jose	48	45	12.7	5.3	2.4	115	218	1422	165	1.32
Riverside	58	49	20.4	9.7	2.1	377	149	3675	146	1.02
Sacramento	60	58	24.0	9.0	2.7	233	314	1745	207	1.52
San Diego	63	59	21.3	8.4	2.5	377	273	3075	201	1.36
San Francisco-Oakland	68	65	21.5	6.1	3.5	1547	472	5935	286	1.65
Los Angeles	80	71	21.2	10.6	2.0	3827	533	12473	316	1.68

[a] MSAs with >1 million population in 1980.

[b] Index of dissimilarity, from Farley and Frey[29]

[c] From 1990 Census reports for each MSA/PMSA; 1990 populations for the San Francisco and Oakland PMSAs were combined, for comparability with the (former) San Francisco-Oakland MSA (1980 Census). Rates are estimates, due to changes in definitions of boundaries of some MSAs between 1980 and 1990.

segregation indexes. Using numbers of deaths for males in 1987-89[34] and the black and white male populations (15-24 to 35-44 year-olds) of each MSA/PMSA for 1987-89 (estimated by linear interpolation between the 1980 and 1990 censuses), age-specific rates (15-24 to 35-44 years) were standardized by the "direct" method, with the distribution of the 1980 U.S. population from age 15-24 to 35-44 as the "standard." The black/white ratio of these age-standardized death rates in California MSAs/PMSAs (Table 3) increased with increasing black-white segregation (i.e., index of dissimilarity). The black/white ratio of poverty rates in these seven MSAs/PMSAs also tended to increase as the segregation index increased, but while Los Angeles had the highest black/white mortality ratio and the highest segregation index, it did not have the highest black/white poverty-rate ratio. Except for Anaheim-Santa Ana and San Jose, poverty rates for blacks in all other MSAs/PMSAs were within a narrow range (20-24%), but mortality rates for blacks varied considerably and tended to increase with increasing level of segregation. Again, the low black/white ratios of age-standardized death rates in several MSAs are noteworthy. Numbers of deaths for black females in these MSAs during 1987-89 were too small for meaningful analyses.

Interpretations

The studies described above involved large geographic units (i.e., cities or MSAs) rather than smaller units (census tracts or block-groups) recommended for ecologic epidemiologic studies involving social class indicators.[35] If the association between residential segregation and mortality by MSA is not due entirely to confounding with social class and the high cost of living in ghettos (as noted even in 1950 for predominantly non-white New York City areas),[26] variation in medical care, psychosocial factors and the physical environment among blacks by degree of segregation could be involved.

In Chicago, as in many other areas, high black-white segregation means that blacks (regardless of income) tend to reside in census tracts with high poverty rates.[18] The finding that even black infants with "adequate" prenatal care and full-term gestation had high neonatal death rates suggested a role for factors such as housing, nutrition, various stresses, and quality of medical care in segregated (predominantly black) areas.[17] Higher infant mortality rates in U.S. blacks

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than whites are due both to higher rates of low birthweight and higher death rates within normal birthweight (but not within low birthweight) infants. The degree of segregation in 176 cities has been associated with black infant mortality rates even when the low birthweight rate was included in the regression model.[31] Additional modeling is needed to evaluate the effect of segregation on infant mortality due to (and independent of) an association with low birthweight. In a study of college-educated parents, the black-white difference in infant mortality was due entirely to a higher prevalence of low birthweight in blacks.[10,11] A study showing high low birthweight rates in biracial infants of black mothers has raised the possibility of adverse effects of stress and racism experienced by black mothers.[36] Further examination is required. Other studies suggest that intensive prenatal care, reduction in stress and control of idiopathic premature labor may be required to reduce the black-white difference in low birthweight rates.[37]

The segregation of blacks in poor, inner-city areas is associated with lack of health insurance and use of overcrowded hospital emergency rooms, which may contribute to uncontrolled hypertension in young, adult blacks in New York City[38] and death from preventable causes (including hypertension) among blacks in the District of Columbia.[4] Medical care access and quality, however, may be related to segregation and discrimination in complex ways, as discussed below.

Discrimination and segregation in relation to health: mechanisms

Discriminatory and racist experiences (defined elsewhere[20]) of blacks could vary with the degree of residential segregation. According to the "social contact" theory, racial prejudice in whites is reduced by protracted contact with blacks (especially if begun at a young age),[20] and greater contact tends to occur in less segregated areas. However, the initial experiences of blacks moving out of segregated areas and seeking upward social mobility by obtaining bank loans for business, may involve discrimination and racism. Moving into less segregated areas could increase perceived incongruity between high socioeconomic status and low social status—due to dark skin color, as discussed below. In this review, the effects of discrimination independent of and interacting with residential segregation cannot be distinguished.

Medical care

The history of the effects of a white-dominated medical system on the health of blacks, including the Tuskegee Syphilis Study and its persistent effects on the black community, has been reviewed elsewhere.[20,39,40] Studies of diagnostic and treatment services for heart disease in adults suggest that black-white disparities persist after adjustment for socioeconomic indicators by census tract of residence.[41,42] The black-white disparity in access to, and waiting time for, kidney transplantation may reflect discrimination on the basis of both social class and race, and the desire for matching on tissue antigens (human leukocyte antigens or HLA) is disadvantageous for blacks. Allocation strategies have been proposed to increase racial equity.[43] After excluding renal diseases, elderly black Medicare enrollees were less likely than whites to receive 23 of 32 medical procedures or tests, even after some adjustment for financial barriers; physician decision-making processes for black patients may be relevant.[44]

In a nationally representative sample of hospitalized Medicare patients, quality of care was lower for patients who were black or came from poor neighborhoods regardless of type of hospital; however, because black patients were often treated at urban teaching hospitals (with better-quality care than rural or urban-nonteaching hospitals), overall quality of care and death rates did

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not differ between blacks and whites.[45] In Veterans Administration hospitals, where care is free and black-white socieconomic differences are smaller than in the general population, the number of major coronary procedures was lower for black than for white patients with acute myocardial infarction, but two-year survival rates did not differ.[46] Studies are needed on survival with medical conditions other than cardiovascular diseases, along with consideration of quality of life after hospitalization.[47] Also, follow-up studies of hospitalized patients do not include deaths among persons never hospitalized; death rates from ill-defined causes are higher in blacks and other minorities than in whites, due in part to lack of access to care,[48] and could vary by degree of residential segregation.

Psychophysiological responses and health

Studies on the physiological effects of experiences with racial discrimination on health have focused on the etiology of hypertension in blacks. In a study of a small sample of women in Alameda County California, Krieger[49] found that, among blacks, acceptance of (or internalized response to) discrimination was associated with higher blood pressure. This is consistent with the finding that the experimental viewing of scenes depicting racist situations was associated with acute increases in blood pressure in a sample of black college students.[50]

"Status incongruence" in blacks, or higher-class lifestyle associated with low social status (due to dark skin color),[51] may be interpreted as resulting from discrimination. The degree of status incongruence was positively associated with blood pressure level and prevalence of hypertension among blacks in a small Alabama city, suggesting that "frequent frustrated social interactions" may lead to "autonomic arousal" and elevated blood pressure.[51] This situation may resemble that shown for demand-control aspects of the work environment in relation to cardiovascular reactivity and hypertension or heart disease.[52] Cooper[53] suggested that the psychological impact of racism might contribute to "conditions ranging from psychiatric illness to hypertension." Studies on self-reported experiences with discrimination in relation to mental health have been reviewed elsewhere.[16]

In addition to black-white differences in exposure to certain stresses, reactions to psychosocial stress in blacks could have health consequences. "John Henryism"[54,55] (named after the black folk figure, John Henry) or hard-driving effort and active coping responses to psychosocial stressors, may influence health in a complex fashion; that is, high levels of John Henryism combined with low social class were associated with a higher prevalence of hypertension. Apparently, the prevalence of high scores on John Henryism has not been examined among blacks living in areas that differ in degree of segregation.

In conclusion, further studies are needed, preferably longitudinal in design, on the health effects of psychophysiological reactions to stress and perceived "status incongruence" among blacks in urban ghettos and elsewhere.

Environmental exposures and community empowerment

Black-white disparities in exposure to hazardous substances in the occupational[56] or general environment, including blacks more frequently residing near hazardous waste disposal sites,[57-59] may have some impact on black-white differences in death rates. Community organizations were apparently effective in altering waste disposal patterns in Houston.[60] More broadly, various structural characteristics of predominantly black communities could be relevant to mortality and morbidity. LaVeist showed that "relative black political power," or the proportion of blacks on the city council divided by the proportion of blacks in the voting age group, was a significant predictor

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of black (postneonatal) infant mortality rates in U.S. cities with at least 50,000 population.[61] Community-organized health centers, often located in predominantly minority inner-city areas, can be both a focus for community-oriented primary health care and a catalyst for various community activities such as improving the physical environment.[62] Schools are another part of a community's infrastructure, and some black educators (beginning, perhaps, with W.E.B. DuBois)[63] have argued that the quality of education of black children, rather than desegregation, should be emphasized; however, desegregation may result in improvement in the academic performance of blacks.[20] The reality that blacks often attend black-majority schools has suggested the need for "forceful multicultural education," as part of a larger effort of black communities to understand and attempt to overcome their problems.[64]

Conclusions

Racism has been called the "missing variable" in studies of racial-ethnic differences in health.[65] This review has shown that ecologic studies of the association between mortality and segregation, which for blacks is largely involuntary and an outcome of racism and discrimination,[20] are needed involving geographic units smaller than cities or metropolitan areas, including analyses of cause of specific mortality (especially, homicides vs. other causes). Also needed are studies aimed at explaining the relatively low mortality of blacks living in certain less-segregated urban areas (especially in California). Longitudinal or cohort epidemiologic studies of blacks should obtain personal histories of social class indicators, experiences with discrimination, psychological reactions to discrimination, along with the segregation indexes and quality of environment of past residential areas, as potential "risk factors" for disease and death. Such studies could benefit from collaboration between epidemiologists and both social and behavioral scientists. Continued study of trends in and factors associated with discrimination and segregation,[20,66] on the physiologic pathways whereby discrimination might influence physical and mental health, on the role of racism in health care,[67] and on the impact of community empowerment on health[61] is important in developing strategies to interrupt the causal paths[53] (Figure 1) by which racism and discrimination may affect the health status of blacks.

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